

Problem-Solution Fit

Date	15 March 2026
Team ID	xxxxxxx
Project Name	EcoTrack – AI-Powered Carbon Footprint Tracker
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS]

- Students
- Working Professionals
- Households
- Environmentally Conscious Individuals
- Smart City & Sustainability Enthusiasts

2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Lack of awareness about daily carbon emissions.
- Difficulty tracking travel, food, and energy usage.
- No personalized sustainability guidance.
- Happens **every day** during routine activities.

3. TRIGGERS TO ACT [TR]

- Rising climate change awareness.
- Need to reduce carbon footprint.
- Interest in sustainable living.
- Government and environmental campaigns.

4. EMOTIONS BEFORE / AFTER [EM]

Before

- Unaware
- Confused
- Concerned
- Guilty about environmental impact

After

- Informed
- Motivated
- Responsible

- Confident in eco-friendly decisions

5. AVAILABLE SOLUTIONS (PROS & CONS) [AS]

Available Solutions

- Online carbon footprint calculators
- Sustainability tracking apps
- Environmental awareness websites

Pros

- Basic carbon calculations
- Easy access
- General environmental information

Cons

- No AI-powered personalized suggestions
- Limited analytics and visualization
- No habit tracking or achievement system

6. CUSTOMER LIMITATIONS [CL]

- Limited environmental knowledge.
- Limited budget for premium applications.
- Dependence on smartphones and internet.
- Lack of time to manually calculate emissions.

7. BEHAVIOR + ITS INTENSITY [BE]

- Frequently use smartphones and digital applications.
- Occasionally monitor environmental impact.
- Low consistency in maintaining eco-friendly habits.
- **Medium to High** intensity of digital usage.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Web Applications
- Social Media
- Environmental Websites
- Educational Platforms

OFFLINE

- Daily Travel
- Household Energy Usage
- Food Consumption
- Colleges
- Workplaces
- Community Awareness Programs

9. PROBLEM ROOT / CAUSE [RC]

- Lack of awareness about personal carbon emissions.
- No simple tool to monitor daily habits.
- Limited access to personalized sustainability advice.
- Difficulty understanding environmental data.

10. YOUR SOLUTION [SL]

EcoTrack is a Generative AI-powered sustainability platform that helps users monitor, analyze, and reduce their personal carbon footprint. It enables users to log daily travel, food, and energy activities, calculates emissions using the Carbon Interface API, securely stores data in Firebase Firestore, provides personalized eco-friendly recommendations using the Groq LLaMA 3.3-70B model, visualizes environmental trends through interactive dashboards and charts, tracks personal carbon budgets, and rewards sustainable behavior through streaks and achievement badges. This content is concise enough to fit inside the SmartBridge Problem-Solution Fit Canvas while accurately reflecting your EcoTrack project.